The Elephant Random Walk Across Its Full Phase Diagram: Exact Moments, Boundary Martingales, and Limit Laws

HCS-C316 theorem-and-evidence package

3 September 2026 — revision round 1

Abstract

We give a boundary-complete account of the one-dimensional elephant random walk for every memory parameter $p \in [0,1]$ and initial bias $q \in [0,1]$. Exact recurrences yield the first two moments; the singular cases $p = 0$ and $p = 3/4$ receive separate martingale and harmonic formulas. The diffusive, critical, and superdiffusive limits are stated with exact constants, and the first four moments of the superdiffusive limit expose the deterministic and two-point endpoints at $p = 1$. A content-addressed exact computation audits finite identities but is explicitly not offered as proof of an asymptotic theorem.

1 Model and finite laws

Let $X_1 \in \{-1,1\}$ satisfy $\Pr(X_1 = 1) = q$. Given $\mathcal{F}_n = \sigma(X_1,\dots,X_n)$, choose K uniformly in $\{1,\dots,n\}$ and set $X_{n+1} = X_K$ with probability p and $X_{n+1} = -X_K$ otherwise. Put

$$S_n = \sum_{j=1}^{n} X_j, \qquad a = 2p-1, \qquad b = 2q-1, \qquad G_n(c) = \prod_{j=1}^{n-1}\left(1+\frac{c}{j}\right).$$

The model originates with Schütz and Trimper [1]; the martingale phase analysis used below follows Bercu [2].

Theorem 1 (Exact kernel and moments)**.** *For every $n \ge 1$,*

$$\Pr(X_{n+1} = 1 \mid \mathcal{F}_n) = \frac{1}{2}\left(1+\frac{aS_n}{n}\right), \qquad \mathbb{E}(X_{n+1} \mid \mathcal{F}_n) = \frac{aS_n}{n}, \tag{1}$$

$$\mathbb{E}S_n = bG_n(a). \tag{2}$$

Furthermore, with $H_n = \sum_{j=1}^{n} j^{-1}$,

$$\mathbb{E}S_n^2 = \begin{cases} \dfrac{2aG_n(2a)-n}{2a-1}, & a \ne \frac{1}{2}, \\ nH_n, & a = \frac{1}{2}. \end{cases} \tag{3}$$

If $p > 0$, $M_n = S_n/G_n(a)$ is a martingale. At $p = 0$ instead, $S_2 = 0$ and $\widetilde{M}_n = (n-1)S_n$, $n \ge 2$, is a martingale.

Proof. Averaging the signed copy over K proves (1). Hence

$$\mathbb{E}(S_{n+1} \mid \mathcal{F}_n) = \left(1+\frac{a}{n}\right)S_n, \quad \mathbb{E}(S_{n+1}^2 \mid \mathcal{F}_n) = \left(1+\frac{2a}{n}\right)S_n^2 + 1.$$

Iteration gives (2); variation of constants gives (3), with the resonant sum equal to H_n when $2a = 1$. For $p > 0$, divide the first recurrence by $G_{n+1}(a) = (1 + a/n)G_n(a)$. At $p = 0$, this product vanishes after its first factor, but direct substitution of $a = -1$ gives $\mathbb{E}[nS_{n+1} \mid \mathcal{F}_n] = (n-1)S_n$; also $X_2 = -X_1$. ☐

2 The complete phase transition

Theorem 2 (Three regimes and endpoints)**.** *The following alternatives exhaust $p \in [0,1]$.*

1. *If $p < 3/4$, then $S_n/\sqrt{n} \Rightarrow N(0,(3-4p)^{-1})$.*
2. *If $p = 3/4$, then $S_n/\sqrt{n\log n} \Rightarrow N(0,1)$.*
3. *If $p > 3/4$, then $S_n/n^{2p-1} \to L$ almost surely and in L^4. Writing $b = 2q-1$,*

$$\mathbb{E}L = \frac{b}{\Gamma(2p)}, \qquad \mathbb{E}L^2 = \frac{1}{(4p-3)\Gamma(4p-2)},$$

$$\mathbb{E}L^3 = \frac{2pb}{(2p-1)(4p-3)\Gamma(6p-3)}, \qquad \mathbb{E}L^4 = \frac{6(8p^2-4p-1)}{(8p-5)(4p-3)^2\Gamma(8p-4)}.$$

At $p = 1$, $S_n = nX_1$ and $L = X_1$: the limit is deterministic for $q \in \{0,1\}$ and two-point for $0 < q < 1$.

Proof architecture. For $p > 0$, center the martingale from Theorem 1. Its increments are bounded by a deterministic multiple of $G_n(a)^{-1}$. The predictable quadratic-variation sum is governed by $\sum_{j\le n} G_j(a)^{-2}$: it has power growth for $a < 1/2$, logarithmic growth for $a = 1/2$, and a finite limit for $a > 1/2$. The martingale central limit theorem (bounded increments give its Lindeberg condition) yields the first two assertions and their constants. The $p = 0$ normalization in Theorem 1 supplies the omitted endpoint of the diffusive case. For $a > 1/2$, fourth-moment summability gives almost-sure and L^4 convergence. Finally, $G_n(a)/n^a \to 1/\Gamma(a + 1)$ and the exact recurrences through order four give the displayed moments; details follow the martingale derivation in [2]. No finite table is substituted for this argument.

What changes at the threshold. The exact identity $\mathbb{E}S_n^2 = nH_n$ already records the logarithm at $p = 3/4$. Above the threshold the random limit retains the initial bias only in odd moments. At $p = 1$ it reduces to the remembered first sign, which is why a blanket claim of nondegeneracy would be false.

Revision certificate: complete phase and endpoint theorem

This round adds the diffusive, critical, and superdiffusive laws, fourth moments, and the $p = 1$ endpoint classification.

References

[1] G. M. Schütz and S. Trimper, "Elephants can always remember," *Phys. Rev. E* 70 (2004), DOI: 10.1103/PhysRevE.70.045101.

[2] B. Bercu, "A martingale approach for the elephant random walk," *J. Phys. A* 51 (2018), DOI: 10.1088/1751-8121/aa95a6.